

What Does Your Model Leave Out?

Executive Brief · Vector Spaces, Bases, Rank and Projections · v1.0

The decision in one sentence

Use a projection to understand how a proposal or observation relates to a chosen model—not to turn mathematical fit into a judgment of value, quality or policy merit.

Why this matters

A model represents observations using a limited collection of directions. A basis describes those directions without redundancy. Rank tells us how many independent directions are available. An orthogonal projection retains the component inside that chosen subspace and places the remainder in a perpendicular residual. The calculation is precise, but its interpretation remains conditional on what was represented, how it was scaled and which distance was used.

$$\text{Observation} = \text{component captured by the model} + \text{residual}$$

A small residual means closeness to this model under this metric. It does not prove that the model includes every relevant factor. A large residual identifies a mismatch that deserves investigation; it does not, by itself, prove that the observation or proposal is defective.

What executives should request

Representation before results

Ask what the rows, columns and coordinates mean, where the selected directions came from, and whether they are independent. Additional columns do not necessarily supply additional information. If coefficients are used to justify action, ask whether they are unique or merely one of several descriptions of the same fitted result.

Evidence beyond a numerical answer

Request the fitted component, the residual, the rank convention and numerical verification. For orthogonal projection, the projector should be symmetric and idempotent, and the residual perpendicular to the model space. These checks establish computational properties, not the suitability of the model for the decision.

Practical rule: never approve a recommendation on the strength of a projection score alone. Require a documented representation, reproducible calculation and explicit interpretation of what remains outside the model.

An illustrative policy example

Suppose three standardized coordinates represent Agriculture, Health and Energy priorities. Two illustrative archetypes—(1, 0.2, 0.6) and (0.2, 1, 0.6)—define the comparison subspace. A proposed profile is (0.9, 0.7, 0.1). All values are teaching examples, not empirical policy estimates or recommended allocations.

Proposed: (0.9, 0.7, 0.1)

Fitted: $(2/3, 7/15, 17/30) \approx (0.667, 0.467, 0.567)$

Residual: $(7/30, 7/30, -7/15) \approx (0.233, 0.233, -0.467)$

The residual is perpendicular to both archetypes. The negative Energy component means that the proposed Energy coordinate is below its fitted component. It is a description of the relationship to the selected archetypes—not evidence that the proposal should be rejected or Energy spending increased.

Three boundaries on interpretation

First, changing the archetypes can change the model space and the conclusion. Their selection needs substantive justification. Second, scaling or weighting coordinates changes the meaning of distance: comparisons must use declared, defensible units and a suitable metric. Third, unconstrained projection does not enforce feasible policy weights.

The example's fitted component uses coefficients $43/72$ and $25/72$. They sum to $17/18$, not one. Linear subspaces generally allow negative coefficients too. Where coefficients must be nonnegative or sum to one, a constrained model is required. Do not assume that a convenient linear-algebra routine automatically respects operational, financial or institutional restrictions.

A better decision statement

"Under these two archetypes and this scaling, the proposal has the reported fitted and residual components. The Energy difference requires explanation. Before recommending a change, we will test alternative defensible representations, confirm feasibility constraints and consider substantive evidence outside this geometric model."

Controls for an accountable analytical workflow

1 • Define and challenge the representation

Document coordinates, units, preprocessing and the reason for the chosen basis or spanning columns. Identify excluded factors. Specify whether the task is description, prediction, compression or a constrained allocation decision; these are not interchangeable objectives.

2 • Verify structure and numerical behavior

Check rank and record tolerance choices when near-dependence matters. Use stable methods such as SVD-based subspace construction and a suitable least-squares solver. Test reconstruction and orthogonality. A successful installation or a visually plausible plot is not mathematical verification.

3 • Review residuals and sensitivity

Display what the model captures and what it omits. Repeat the analysis under plausible changes to scaling, archetypes or tolerance. Report conclusions that change materially. Distinguish uncertainty about numerical computation from uncertainty about whether the model represents the real problem.

4 • Separate technical evidence from authority to act

Record who owns the analysis, who reviewed it, which version was used and what decision is authorized. Preserve the notebook, dependency version and checksums so another person can reproduce the calculation. Do not treat successful computation as independent review or as permission to publish.

Transfer to statistics and AI

Least squares projects observations onto a design-column space; fitted values can be unique even when coefficients are not. PCA projects centered data onto selected orthonormal directions. In attention systems, however, a learned “projection” may be a general linear map and need not satisfy the mathematical properties of an orthogonal projector. Ask for the actual definition, not just the label.

SRAI control note

Prepared from the owner-approved Lesson 5 chapter and notebook v0.2.2. This brief is a new production asset for review, not independent sign-off or public-release authorization. The pinned `srai_math` runtime is 1.1.1rc1, a prerelease. The companion chapter and exercises provide the derivations and worked checks behind this brief.